

Dimensionality Reduction for Face Recognition

PCA Eigenfaces vs. Autoencoder

Technical Analysis Report

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1. Introduction

This report presents a comparative study of two dimensionality reduction techniques applied to face image data from the ORL (AT&T) face database: Principal Component Analysis (PCA) with eigenfaces and a neural autoencoder. Both methods compress high-dimensional pixel data into compact representations, enabling downstream tasks such as face recognition, clustering, and reconstruction.

The ORL dataset consists of 400 grayscale images of 40 subjects (10 images each), each of size 92×112 pixels, yielding a raw feature dimension of 10,304. Both models were trained on a subset of this data and evaluated on their ability to reconstruct unseen faces.

2. Methods

2.1 PCA — Eigenfaces

PCA identifies the principal axes of variation in the face dataset by performing an eigendecomposition of the covariance matrix. Because $n_{\text{features}} \gg n_{\text{samples}}$, a compact covariance trick is employed: the surrogate matrix $L = X_c \cdot X_c^T$ of shape $(n_{\text{samples}} \times n_{\text{samples}})$ is decomposed instead, and eigenvectors are mapped back to the full feature space. This dramatically reduces computational cost.

The number of components retained is governed by a variance threshold $\alpha \in \{0.80, 0.85, 0.90, 0.95\}$, which determines how many principal components are necessary to explain that fraction of the total variance. The resulting eigenfaces form an orthonormal basis for the face subspace.

2.2 Autoencoder

The autoencoder is a fully-connected neural network with symmetric encoder and decoder branches. The encoder maps the 10,304-dimensional input through layers of $1024 \rightarrow 512 \rightarrow 128 \rightarrow 64$ neurons (with BatchNorm and ReLU activations) to a 64-dimensional latent code. The decoder mirrors this structure, ending with a Sigmoid activation to constrain pixel values to $[0, 1]$.

Training minimises Mean Squared Error (MSE) between input and reconstruction. An Adam optimiser with a step-based learning-rate schedule (halved every 100 epochs) is used. The model is trained from scratch when no checkpoint exists, and the trained weights are saved for reproducibility.

3. Eigenface Visualisation

The figures below show the top principal components (eigenfaces) retained for each variance threshold. Each component captures a distinct pattern of variation across the training faces — early components encode coarse lighting and pose, while later components capture finer details like glasses or facial hair.



Figure 1: Top eigenfaces — 80% variance threshold (~31 components)



Figure 2: Top eigenfaces — 85% variance threshold (~43 components)



Figure 3: Top eigenfaces — 90% variance threshold (~65 components)



Figure 4: Top eigenfaces — 95% variance threshold (~100+ components)

As the variance threshold increases, more components are retained. The higher-order eigenfaces (right side of each row) become increasingly noisy and texture-like, capturing residual variance that corresponds to fine-grained facial details.

4. PCA Reconstructions

Each row below shows how well PCA reconstructs held-out face images at different variance thresholds. Images are sorted from the best-reconstructed (leftmost) to the most challenging (rightmost). The progressive improvement in sharpness and detail is evident as more variance is retained.



Figure 5: PCA reconstructions — 80% variance (fewest components, highest compression)



Figure 6: PCA reconstructions — 90% variance



Figure 7: PCA reconstructions — 85% variance



Figure 8: PCA reconstructions — 95% variance (most components, lowest compression)

At $\alpha = 0.80$ the reconstructions are soft and ghost-like, with major facial structure visible but fine details lost. By $\alpha = 0.95$ the reconstructions closely match the originals, preserving glasses, beard texture, and expression nuance.

5. Autoencoder Reconstructions

The autoencoder was evaluated on 8 held-out images. Each column shows an original (top) paired with its autoencoder reconstruction (bottom). Unlike PCA, the autoencoder learns a nonlinear manifold and can represent complex variation that linear projections miss.



Figure 9: Autoencoder — original (top row) vs. reconstructed (bottom row) for 8 test images

The autoencoder reconstructions are notably sharp and faithful, preserving pose, expression, and accessories such as glasses. Even challenging angles (e.g. subject #158, looking up) are handled well. Minor smoothing is observed in fine hair detail, a common characteristic of MSE-trained autoencoders.

6. Comparative Summary

The table below summarises key characteristics of each configuration:

Model	Variance Threshold / Config	Components / Latent Dim	Reconstruction Quality
PCA	80%	~31 components	Basic — visible noise
PCA	85%	~43 components	Moderate detail
PCA	90%	~65 components	Good — minor blur
PCA	95%	~100+ components	High — fine features
Autoencoder	Latent dim = 64	64 neurons	Excellent — sharp & clear

Key observations:

1. PCA is fast, interpretable, and requires no training. Its components have a clear geometric meaning (variance directions), making it auditable and suitable for resource-constrained settings.
2. The autoencoder produces superior reconstructions at an equivalent or smaller latent dimension (64 vs. 100+ for PCA at 95%), thanks to its nonlinear representation capacity.
3. PCA reconstruction quality degrades gracefully as the threshold decreases, making it easy to trade accuracy for compression.
4. The autoencoder is more expensive to train but offers a better reconstruction floor and can generalise more robustly to unseen variations.

7. Conclusion

Both PCA eigenfaces and the autoencoder successfully compress face images into compact codes suitable for downstream recognition tasks. PCA at 95% variance is recommended when interpretability and training speed are priorities, while the autoencoder is preferred when reconstruction fidelity is critical. A combined approach — using PCA for initialisation and the autoencoder for fine-tuning — could yield further benefits in practice.

Future work may explore variational autoencoders (VAEs) for probabilistic latent spaces, convolutional architectures for better spatial feature capture, and contrastive learning objectives for improved face verification performance.